

# Evaluating Masked Language Model Understanding of Finnish Cases

Cody Daniels

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Submitted to Professor Dougal

Department of Linguistics

Brigham Young University

codymd@byu.edu

## Abstract

This study evaluates how well masked language models understand Finnish case morphology, a notoriously complex aspect of the language. Using FinBERT and XLM-RoBERTa, I assessed model performance by comparing the log probabilities of correct and synthetically-generated incorrect case forms in natural contexts from the Yleisradio Finnish news corpus. Focusing on commonly confused case pairs—such as partitive vs. accusative and various internal vs. external locative cases—I measured the models’ ability to prefer the grammatically correct form. FinBERT consistently outperformed XLM-RoBERTa, achieving over 90% accuracy in most experiments and demonstrating greater confidence in its predictions. These findings suggest FinBERT’s suitability for educational applications involving Finnish morphology, while also highlighting multilingual models’ challenges with representing low-resource grammatical features. Future work could explore verb rection contexts to further test model understanding of syntactic dependencies. *Abstract generated with assistance from ChatGPT (OpenAI, 2024).*

## 1 Introduction

Large language models (LLMs) have consumed tons of language data in their training and thus contain a lot of valuable information that might prove useful to a second-language learner. For learners of Finnish, one of the most difficult aspects of the language to learn is its system of 15 cases. A common question to ask might be, “Am I using the right case?” A masked language model could theoretically be used to evaluate several candidate cases and determine which is the most likely given the bidirectional context (Devlin et al., 2019). FinBERT and multilingual models like XLM-RoBERTa have been previously evaluated on traditional NLP tasks like document clas-

sification, named entity recognition, and part-of-speech tagging, but not directly on their knowledge of Finnish’s morphological features (Conneau et al., 2019; Virtanen et al., 2019). In this project, I evaluated the accuracy of FinBERT and XLM-RoBERTa predicting the correct grammatical case in a variety of contexts.

## 2 Literature Review

Perhaps the most relevant piece of previous research is a recent paper by Moisio et al. which evaluates generative language models’ ability to analyze the morphology of complex forms of Finnish nouns as produced by finite-state transducers (FSTs) (Moisio et al., 2024). Their results show that large models like GPT-4-turbo and GPT-3.5-turbo struggle while smaller models like Llama2-70B and Finnish models like Poro-34B fail significantly. Only GPT-4-turbo is able to approach the accuracy of a simple recurrent neural network (RNN) for identifying the correct number, case, or possessive suffix of a complex noun form. The results suggest that these models struggle to develop a real understanding of Finnish grammar. My experiment differs from Moisio et al. by evaluating masked instead of generative language models. My experiment also evaluates the correct use of natural forms in context rather than context-free FST-generated forms.

For this paper, I use masked language models that are based on the BERT-architecture (Devlin et al., 2019). BERT stands for Bidirectional Encoder Representations from Transformers. It uses self-attention mechanisms to jointly condition on both left and right context in all layers. The bidirectionality of BERT is crucial for the task of case disambiguation since pertinent information may come from before or after its use. The more recent RoBERTa model enhanced the original BERT pretraining approach and matched or exceeded BERT’s performance on common bench-

marks (Liu et al., 2019). XLM-RoBERTa, trained on 100 languages using the RoBERTa paradigm, became the go-to masked language model for non-English NLP and is thus evaluated in my experiments (Conneau et al., 2019). TurkuNLP’s FinBERT is the predominant masked language model for Finnish-centric tasks and is included in my experiments as well (Virtanen et al., 2019).

### 3 Methodology

#### 3.1 Experimental Design

Masked language model understanding of Finnish cases was measured by comparing the probabilities of two forms of the same word (one with the correct case and an artificially-generated form with an incorrect case) in a naturally-occurring context. The incorrect form is a commonly confused form so as to maximize the difficulty of the task. If the correct case is given a higher joint probability, then the model passes that test case. Joint probability is calculated by multiplying together the probabilities of each token that makes up the word. This is done since putting a different case on a word may sometimes require a different number of tokens. In practice, log probabilities are used, so they are simply added together.

There are certain pairs of grammatical cases that are especially tricky for Finnish-as-a-foreign-language learners to distinguish, so I focused on those. Experiments were run in both directions for each pair. The pairs are the following:

1. Partitive vs. Accusative
2. Illative vs. Allative
3. Elative vs. Ablative
4. Inessive vs. Adessive
5. Translative vs. Essive

The use of partitive vs. accusative is especially difficult to distinguish since they are both object cases, and partiality is not a distinction that is present in many other world languages. Pairs 2 through 4 are all locative cases, meaning they describe location. Each of these pairs contain an internal and an external case. The last pair contains the two essive cases, meaning they describe existence.

#### 3.2 Data Collection

Since both FinBERT and XLM-RoBERTa were released in 2019, I needed data from post-2019 to make certain that it was not included in the training data. The Finnish national broadcasting service (Yleisradio) provides a public dataset of Finnish language news articles in a treebank format from 2019 to 2021 that meets this requirement (Yle, 2019-2021). I decided to use all the data from the year 2020.

The Yleisradio dataset contained many extraneous sentences. The sentences were filtered to meet the following criteria:

1. The sentence must contain a word with the target case.
2. The target word must have no dependencies.
3. The target word must not have case agreement with its parent.
4. The target word must not have clitics or possessive suffixes.
5. The target word must be present in the dictionary.

Requirements 2 and 3 were instituted to simplify the process of replacing its form with an incorrect one. Any case agreement would require more sentence modification and more complex masking for evaluation. Requirement 4 simplifies the alternative form generation. Requirement 5 is necessary since alternative forms are generated by looking up the lemma in a dictionary with declension data. The dictionary used for this project is Wiktionary, since it is freely available and contains comprehensive declension guides (Wiktionary contributors, 2025). The `voikko` Python package provided easy access to this declension data (Pitkänen and Hauhio, 2018).

In total, I gathered 12,500 examples for each experiment direction with the exception of accusative to partitive. Due to the complexity of how the accusative case appears in Finnish, only pronouns were denoted as accusative in the treebank data. As a result, only 7,100 accusative examples were found after parsing all of the treebank data. The partitive to accusative experiment also only uses pronouns due to the complexity of generating the accusative form of an arbitrary noun.

As previously stated, each example context was evaluated using the correct and incorrect form with

| Model                                | Internal Locative Cases |               |               | External Locative Cases |               |               | Essive Cases  |               | Grammatical Cases |               |
|--------------------------------------|-------------------------|---------------|---------------|-------------------------|---------------|---------------|---------------|---------------|-------------------|---------------|
|                                      | Ela to Abl              | Ine to Ade    | Ill to All    | Abl to Ela              | Ade to Ine    | All to Ill    | Ess to Trans  | Trans to Ess  | Part to Acc       | Acc to Part   |
| TurkuNLP/bert-large-finnish-cased-v1 | <b>97.38%</b>           | <b>95.32%</b> | <b>93.98%</b> | <b>83.90%</b>           | <b>88.34%</b> | <b>90.49%</b> | <b>97.24%</b> | <b>94.55%</b> | <b>95.93%</b>     | <b>94.93%</b> |
| TurkuNLP/bert-base-finnish-cased-v1  | 97.34%                  | 94.82%        | 92.98%        | 81.84%                  | 86.63%        | 89.66%        | 96.84%        | 94.04%        | 95.41%            | 94.49%        |
| FacebookAI/xlm-roberta-large         | 92.50%                  | 84.25%        | 75.43%        | 51.48%                  | 57.90%        | 69.89%        | 80.55%        | 75.46%        | 92.59%            | 56.99%        |
| FacebookAI/xlm-roberta-base          | 90.14%                  | 79.32%        | 71.22%        | 44.44%*                 | 56.88%        | 63.88%        | 75.04%        | 70.46%        | 85.14%            | 50.73%*       |

Table 1: Percent accuracy of identifying the correct case given a natural context. \*Statistically insignificant compared to random guessing.

FinBERT and XLM-RoBERTa. Using these two models elucidates the effects of tokenization and training data on Finnish case understanding. Both base and large sizes of each model were used to determine the effect of model size on understanding as well.

## 4 Results

### 4.1 Model Accuracy

Across all experiments, FinBERT models outperformed XLM-RoBERTa models (see Table 1). Furthermore, FinBERT obtained impressive accuracy with greater than 90% of examples correctly identified in all experiments except for those from external locative cases. This shortcoming is perhaps due to the imbalance in frequency between internal and external locative cases. Interval locative cases are far more prevalent and have more versatile extended uses as well. Both models show a statistically significant difference from random guessing (50% accuracy) according to a one-tailed one-sample binomial test. The base model of XLM-RoBERTa, however, shows no significant difference for the ablative to elative and accusative to partitive experiments.

Increasing model size seems to yield only a negligible improvement for FinBERT (under 2 percentage points). The large version of XLM-RoBERTa shows some more sizable improvements over its base model. This improvement suggests that a larger model is necessary to store important grammatical information for all of its 100 languages due to the limited number of parameters per language. Since FinBERT is Finnish-centric, it can store the pertinent information in a smaller footprint.

Some noise is to be expected in these results, since both cases (the naturally-occurring one and the synthetically-produced one) may be correct in certain contexts. In these instances, swapping cases just changes the meaning of the sentence rather than being a grammatical error. So, for example, it may not be possible to achieve an accu-

racy much greater than 97.38% (FinBERT’s best result).

### 4.2 Model Confidence

Figure 1 shows the difference in log probabilities between the correct case and the synthetic case for each instance in the partitive vs. accusative experiment (both directions). Partitive vs. accusative is one of the most common and important distinctions to get right in Finnish. The FinBERT model is rarely confidently wrong in this experiment. In other words, there is a short tail on the left side of the red line where incorrectly identified examples lie. XLM-RoBERTa, on the other hand, is often confidently wrong. This conclusion is evident from the very long tail on the left side and even a second mode around -10 which suggests systematic errors. For FinBERT, the bulk of the probability mass is safely on the right side of the line, while the probability mass for XLM-RoBERTa is much closer to the line. This suggests that XLM-RoBERTa is even less confident about its correct answers as well.

## 5 Discussion

Overall, the results are very promising. Compared to the results of Moisio et al., FinBERT demonstrates more understanding of Finnish cases and morphology in these experiments than their prompting of generative language models with isolated forms (Moisio et al., 2024). Context is extremely helpful to these models. While my evaluation data was not used in the training of FinBERT, it did contain normal and natural noun forms that it was more likely to have already been exposed to. Perhaps FinBERT would struggle with the more novel, synthetic forms they generated with their finite-state transducer.

The high accuracy of FinBERT across all experiments is sufficiently trustworthy to use in educational applications. The analysis of log probabilities shows that output distributions can be used as a reliable proxy for answer confidence, at least for

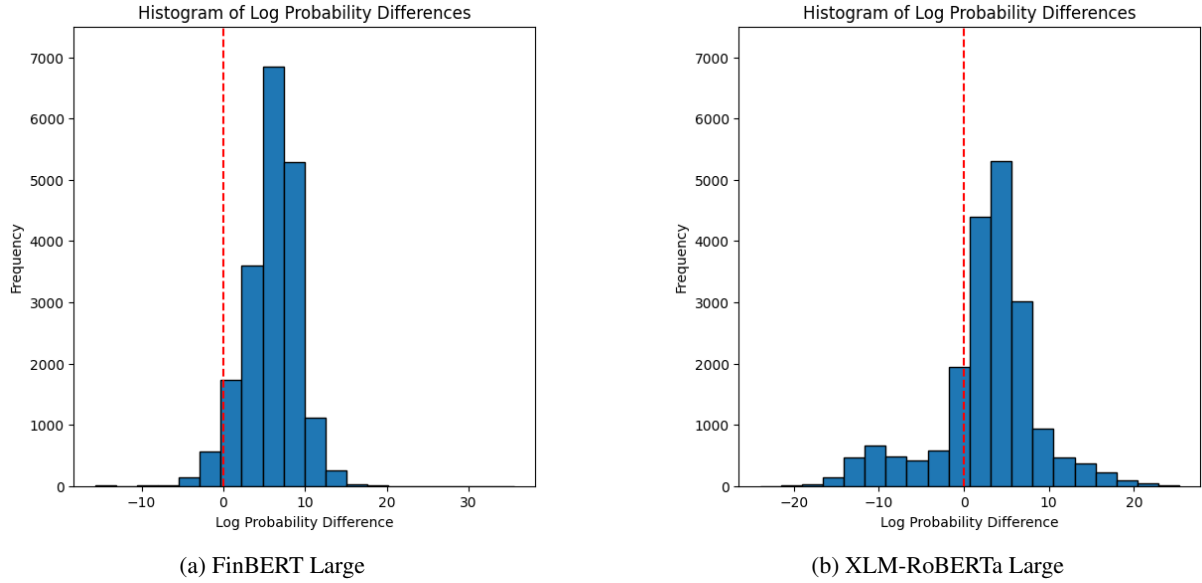


Figure 1: Log Probability Histograms

FinBERT. XLM-RoBERTa, on the other hand, is likely not accurate enough for educational applications where students rely on correct feedback to reinforce their learning. The students themselves are not yet able to distinguish between right and wrong and end up trusting the models to tell them the truth. Appending a confidence score to answers could help students put the model’s answers in perspective.

There are several reasons why FinBERT may have achieved superior performance compared to XLM-RoBERTa. As the original authors of FinBERT describe, multilingual models are not enough for overcoming the challenges that low-resource languages face (Virtanen et al., 2019). Multilingual models’ representations of low-resource languages are lacking when compared to dedicated language models for the same languages. Another benefit of using FinBERT over XLM-RoBERTa is a more intuitive and efficient tokenization scheme. XLM-RoBERTa requires more tokens to encode the same content.

## 6 Conclusion

In this paper, I find that FinBERT shows impressive understanding of the use of Finnish cases with XLM-RoBERTa lagging somewhat behind. The results support the findings of Virtanen et al. that monolingual masked languages models demonstrate superior understanding of low-resource languages compared to multilingual models that try to take advantage of transfer learning (Virtanen et al.,

2019). FinBERT has great potential to be used in educational applications which help Finnish learners distinguish between which case to use.

### 6.1 Limitations

My study made numerous simplifications. The examples that were gathered from the Yleisradio corpus did not contain clitics, possessive suffixes, or case agreement. The news data itself was very much in a familiar domain for the FinBERT base model. Each experiment was represented as a binary choice when it may have been more appropriate to compare the log probabilities for every possible case at once. When framed as a ranking task, a more sophisticated evaluation metric could be used. I chose just five pairs of cases which cover only 10 of the 15 Finnish cases. Nominative, genitive, instructive, abessive, and comitative cases were excluded. There was no mechanism to determine if the synthetic form was actually incorrect or still valid but semantically different.

### 6.2 Future Work

The most immediate and beneficial continuation of these findings is the construction of an educational application for Finnish-as-a-foreign-language learners to build their confidence in selecting the correct case in their speech. Such a tool could either proofread learner-provided texts for accuracy or be a quiz application that generates examples and asks users to verify the correct choice like flashcards.

Another direction to explore would be to evaluate instances where verbs require the use of a certain case. These are known as rections. *Tarkista Tästä* is dictionary of Finnish rections that could be used to find important verbs which demand a certain case (Jönsson-Korhola and White, 1997).

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## A Source Code

The code for this project can be found in a public Google Colab notebook available [here](#).